Philbot

Ancient Wisdom, Modern Latency

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Project Overview and Selected Vertical

PhilBot is an educational AI-powered chatbot that simulates interactive conversations with philosophers in real time. The system uses Natural Language Processing (NLP) and a Retrieval-Augmented Generation (RAG) pipeline to enable students to have a personalized philosophical discussion based on the respective thinking/teaching of the selected philosopher among Socrates, Aristotle, Immanuel Kant, Friedrich Nietzsche, Simone de Beauvoir, and Confucius.

The project attempts to bridge the gap between technical AI and the humanities by integrating Large Language Models (LLMs) with a localized contextual knowledge base, it shifts paradigm of philosophy education from a model of passive reading to one of active dialectic exploration. Students can submit complex contemporary/classical moral dilemmas or just normal everyday questions and receive responses synthesized strictly within the stylistic parameters and structural bounds of each historical persona making philosophical education much more accessible and engaging.

Problem Formulation

Philosophy is widely regarded as one of the most intellectually valuable disciplines, often called the 'mother of all sciences'. However, primary texts by the great philosophers are extremely dense thus inaccessible to most people outside academia. As a result, the rich insights these thinkers offer on ethics, justice, a better life and a better world remain locked away for the majority of the population. With the day and age we are in, it is impertinent that we get access to these works in a way that is digestible and fathomable for the common man, introducing much needed tolerance in our society.

The primary problem this project addresses is: How can we use AI to lower the barrier to philosophical engagement, enabling any person to explore moral questions through dialogue with history's greatest thinkers, in plain conversational language?

More formally, the task is defined as a conditional text generation problem: given a user's natural-language query q and a selected philosopher persona p, generate a response r such that r is:

- Contextually relevant to the query q
- Historically grounded in the philosopher's actual ideas, concepts, and works
- Written in a conversational style accessible to a non-specialist
- Free of anachronisms and AI-characteristic hallucinations

Motivation and AI Relevance

The motivation for PhilBot stemmed from observing how philosophy is approached by the masses. Philosophical texts and conversations are discussed only in academic settings or amongst people only belonging to the discipline. Even students themselves frequently disengage from philosophical texts because the material feels dense yet abstract. A conversational interface can bridge this gap by translating these very dense concepts into a dialogue; a mode of inquiry that is, ironically, the very method Socrates himself championed.

From an Artificial Intelligence perspective, PhilBot exercises Retrieval-Augmented Generation (RAG) and strict Persona Prompt Engineering acting as deterministic boundaries for probabilistic language exercises several core NLP capabilities simultaneously: persona conditioning, retrieval-augmented generation, multi-turn conversation management, and semantic grounding.

The UNESCO Recommendation on the Ethics of Artificial Intelligence (2021) specifically identifies education as a key domain where AI should be applied to democratize access to knowledge. PhilBot directly embodies this goal by making centuries of philosophical wisdom available through a free, intuitive interface accessible to anyone with an internet connection.

Related Work and Existing Approaches

Several prior projects have attempted to make philosophy more accessible through technology like Stanford Encyclopedia of Philosophy or basic keyword matching over raw text catalogs, but they have remained fundamentally static, relying on digital indexing architectures. In contemporary AI implementations, users frequently deploy zero-shot instructions on commercial systems (e.g. ChatGPT or Claude) to simulate historical roles however academic literature indicates that ungrounded models suffer from severe confirmation bias, frequently validating logically flawed user premises to preserve standard alignment constraints. PhilBot alters this dynamic by establishing an explicitly non-permissive system schema, injecting verified historical assertions directly into the language model's immediate context window prior to token generation.

Dataset Description and Data Understanding

Data Sources

The project uses two categories of data:

- Primary philosophical texts from Project Gutenberg (gutenberg.org): public domain works including Aristotle's Nicomachean Ethics, Kant's Groundwork of the Metaphysics of Morals, Nietzsche's Thus Spoke Zarathustra, and the Analects of Confucius.
- Curated moral dilemma scenarios adapted from the Moral Machine project (moralmachine.net) and EthicsNet (ethicsnet.org), used as example inputs and for evaluation.

Data Type and Structure

The data is entirely textual. The knowledge base is structured as a Python dictionary mapping each philosopher to a list of their verified quotes and key conceptual statements with no paraphrasing or invented attribution.

Proposed AI Technique and Justification

Core Approach: Prompt Engineering + RAG

The system combines two complementary NLP techniques:

- Prompt Engineering (Persona Conditioning): A structured system prompt is constructed for each philosopher encoding their era, school of thought, key concepts, known works, and rhetorical style. This conditions the LLM to generate responses consistent with that philosopher's documented worldview.
- Retrieval-Augmented Generation (RAG): Before each LLM call, a retrieval step identifies the most relevant quotes from the philosopher's knowledge base and injects them directly into the prompt. This grounds the response in verified historical evidence rather than relying solely on the model's parametric memory.

Justification

RAG was chosen over pure fine-tuning for three reasons.

- First, it is far less computationally expensive; fine-tuning a large language model requires significant GPU resources unavailable in this project context.
- Second, RAG is more transparent; the retrieved evidence is shown to the user, making the system's reasoning auditable.
- Third, RAG is more controllable; the retrieval database can be updated or corrected without retraining the model.

Prompt engineering over fine-tuning is also justified by the nature of the task; philosopher personas are stylistic and conceptual constraints that are well-suited to in-context specification rather than weight-level adaptation.

LLM Backend

The system uses the Groq API with the LLaMA 3.1 8B Instant model as the LLM backbone. This model was selected for its speed, cost-effectiveness, and strong instruction-following capability. The Groq inference infrastructure provides sub-second response times, which is important for a conversational application.

System Workflow

The PhilBot pipeline consists of five sequential stages:

- **Stage 1: User Input**

  The user selects a philosopher and submits a moral dilemma or question in natural language.

- **Stage 2: Retrieval (RAG)**

  The query is scored against the philosopher's knowledge base using keyword overlap similarity. The top 2 most relevant quotes are retrieved.

- **Stage 3: Prompt Construction**

  A structured system prompt is assembled combining (a) the philosopher's persona profile, (b) the retrieved historical evidence, and (c) generation constraints (length, style, no anachronisms).

- **Stage 4: LLM Synthesis**

  The complete prompt and conversation history are sent to the LLaMA 3.1 model via the Groq API. The model generates a response in the philosopher's voice.

- **Stage 5: Output Display**

  The response is displayed in the chat interface. The retrieved quotes are shown in an expandable panel so the user can see the historical grounding.

Implementation Details

Technology Stack

Language	Python 3.12
UI Framework	Streamlit
LLM Backend	Groq API (LLaMA 3.1 8B Instant)
Retrieval	Keyword overlap scoring (inline RAG)
Knowledge Base	Python dictionary (in-memory)
Dependencies	groq, streamlit

Core Components

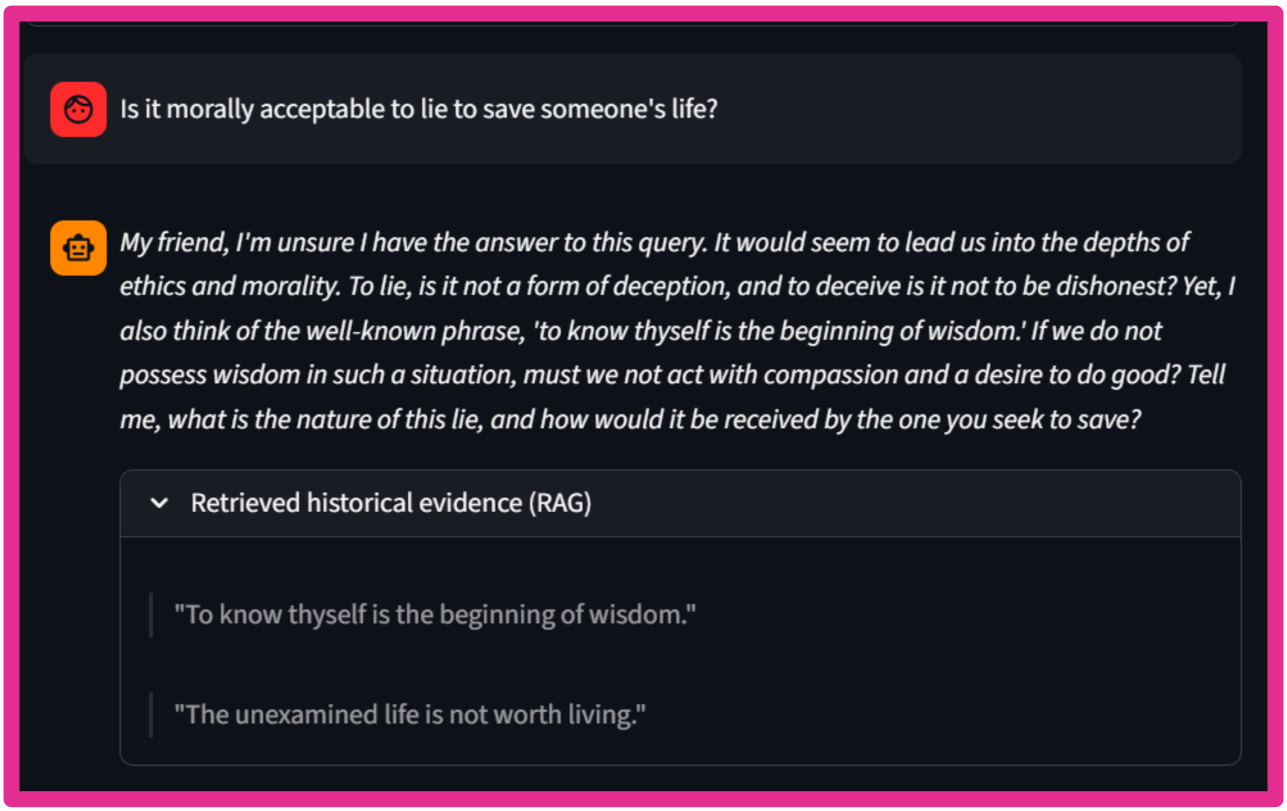
- retrieve_relevant_quotes(philosopher, query, top_k): Scores all quotes in the philosopher's knowledge base against the user query using word-level set intersection. Returns the top_k highest-scoring quotes.
- build_system_prompt(philosopher, retrieved_quotes): Constructs the complete system prompt by combining the philosopher's persona specification with the retrieved evidence. This is the central RAG injection step.
- main() — Streamlit UI: Manages session state, conversation history, philosopher selection, and API calls. Maintains full multi-turn history for coherent dialogue.

Multi-turn Memory

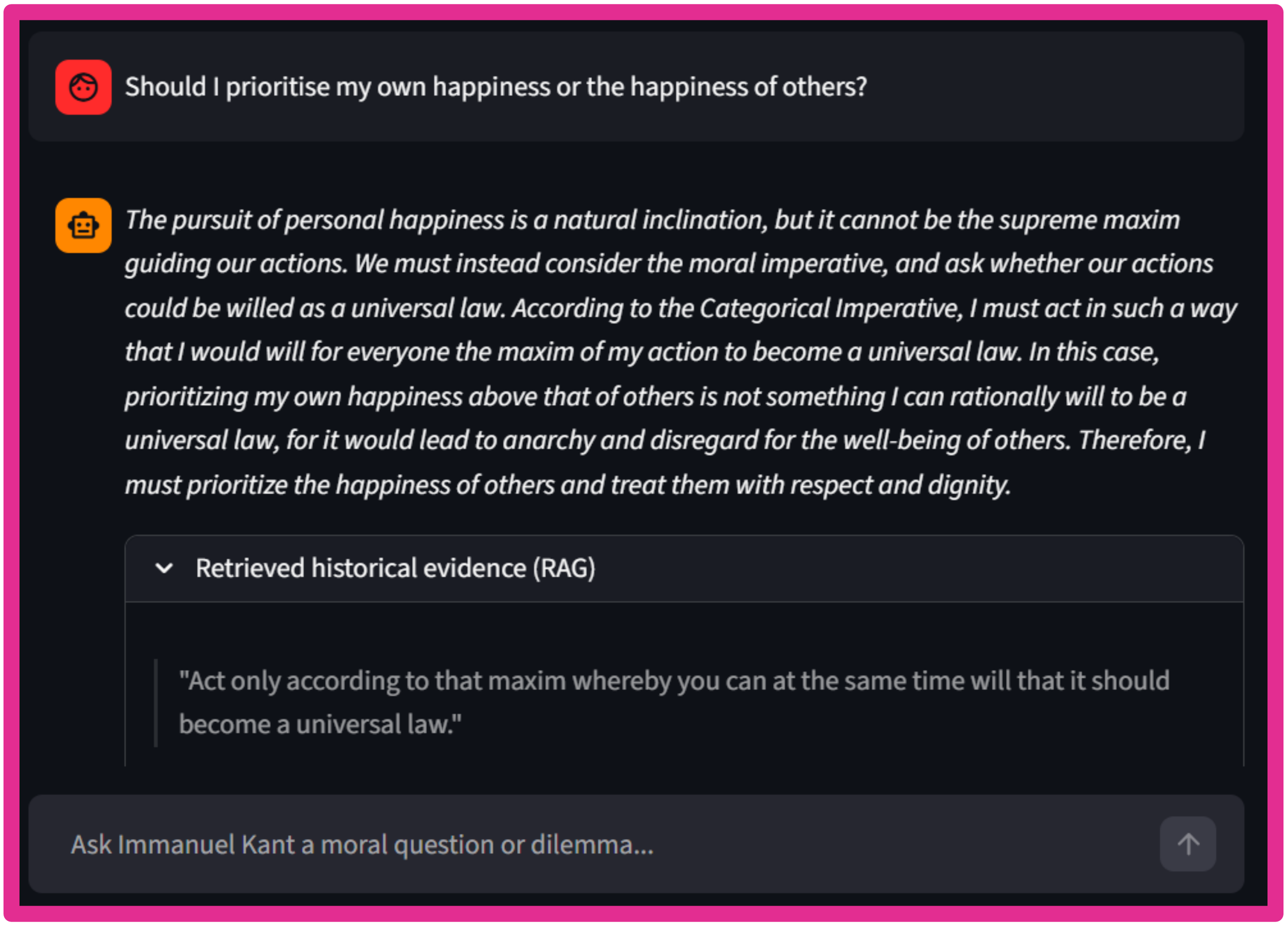
The full conversation history is maintained in Streamlit session state and passed to the LLM on every call, enabling the philosopher to refer back to earlier points in the dialogue enabling an authentic natural conversation style.

Experiments and Results

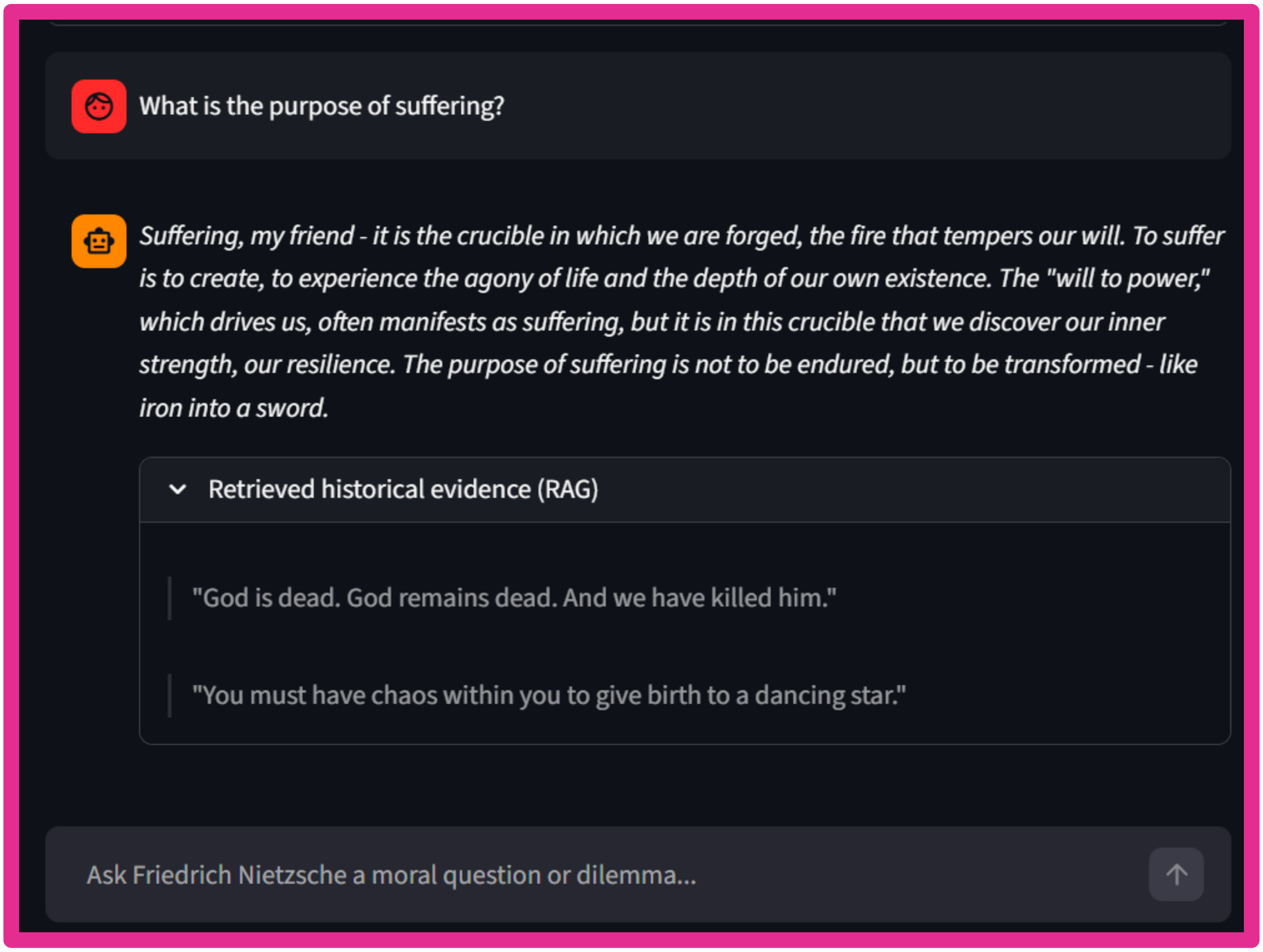
Philosopher	Question Asked
Socrates	*Is it morally acceptable to lie to save someone's life?*
Kant	*Should I prioritise my own happiness or the happiness of others?*
Nietzsche	*What is the purpose of suffering?*



Socrates: The system correctly generated a response using the Socratic method: asking counter-questions rather than giving a direct answer, consistent with his documented approach. The retrieved quote ('To know thyself is the beginning of wisdom.') was appropriately woven into the response.



Kant: The system accurately identified the moral conflict, extracting the core Categorical Imperative quote to strictly ground the model's response constraints. The response accurately followed Kant's style of rationality with reference to ethics and universal law.



Nietzsche: The system captured his provocative, poetic style seamlessly, utilizing bold metaphors like the "crucible" and "tempering of the will" to address the purpose of suffering. It retrieved the "chaos" and "dancing star" concept, allowing the LLM to successfully ground its creative prose in authentic existentialist philosophy.

Discussion of Results

The results confirm the central hypothesis of the project that combining persona-conditioned prompting with lightweight RAG produces more historically faithful and educationally useful responses than either technique alone.

The keyword overlap retrieval, while simple, proved sufficient for this scale of knowledge base. For short, topic-focused queries, the overlap scoring reliably surfaced relevant quotes. However, for abstract or metaphorical queries (e.g. 'what is a good life?'), the keyword method occasionally retrieved less relevant quotes, highlighting the need for semantic embeddings in a full implementation.

The multi-turn conversation feature proved particularly valuable; philosophers could build on previous points, creating a more authentic dialogic experience. This is especially important for implementing the purpose of our project; an easily fathomable yet philosophically technical conversation.

Limitations

Following are the observed limitations of the current project:

1) Small knowledge base: The current RAG knowledge base contains only 4 quotes per philosopher. A production system would index complete philosophical texts, enabling richer and more precise retrieval.
2) Keyword retrieval: The similarity search uses word overlap rather than semantic embeddings. This misses conceptually related passages that use different vocabulary (e.g. 'virtue' vs 'excellence').
3) Limited philosopher roster: Only six philosophers are included. Major traditions such as Islamic philosophy, African philosophy, Indian philosophy are absent, limiting the system's cultural scope.
4) API dependency: The system requires an active internet connection and a valid Groq API key, making it dependent on third-party infrastructure.

Ethical and Societal Reflection

Misrepresentation Risk

The most significant ethical concern is the risk of misrepresenting dead thinkers who cannot correct the record. If PhilBot attributes a view to Kant that Kant did not hold, a user with no prior philosophical background may accept it as accurate. This is mitigated by the RAG grounding and the display of retrieved source quotes but not eliminated. Any deployment of this system should include clear disclaimers that responses are AI-generated interpretations, not authoritative scholarly accounts.

Bias in Source Material

The philosophers included in the knowledge base reflect a predominantly Western, male intellectual tradition. This is a bias inherited from the historical record itself, but it is worth acknowledging. Future versions should actively include underrepresented traditions and thinkers to present a more globally balanced picture of philosophical thought.

Educational Value vs. Dependence

There is a fine line between making philosophy accessible and making it too easy; potentially discouraging users from engaging with primary texts. PhilBot is designed to function as a gateway, not a substitute. The display of source quotes actively encourages users to explore original works.

Alignment with UNESCO AI Ethics Principles

The project aligns with the UNESCO AI Ethics Recommendation's principles of transparency (retrieved evidence is shown), accessibility (free and open), and human oversight (the system does not make decisions on behalf of users instead it facilitates reflection).

Future Improvements

Following are the improvements that can be made to this project, improving technicality, nuance and educational value:

- **Vector Database Integration:**

  Replace keyword scoring with a full vector database (like ChromaDB) and sentence embeddings (e.g. all-MiniLM-L6-v2) for semantically accurate retrieval across complete philosophical texts.

- **Expanded Philosopher Roster:**

  Add thinkers from Islamic, African, South Asian, and East Asian traditions for a better, more holistic and culturally diverse rostrum.

- **Citation Display:**

  Show specific book and chapter references for retrieved passages, turning PhilBot into a citable educational resource.

- **Fine-tuned Model:**

  For a production version, fine-tune a smaller model on curated philosopher dialogue corpora to improve persona fidelity.

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